STUDENT MENTAL HEALTH STATUS PREDICTION

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Abstract : Student mental health is a growing concern that negatively impacts academic performance, social interactions, and overall well-being. Currently, identifying stress and mental health risks relies heavily on manual observation or delayed self-reporting methods, which often lack accuracy and real-time intervention capabilities. To address this gap, data-driven approaches have proven to be effective solutions. This study presents a machine learning-based prediction system designed to classify student mental health status into three actionable categories: Normal, Stressed, and At Risk. Utilizing a comprehensive dataset of 1,100 student records, the system analyzes 20 diverse attributes spanning psychological, physiological, environmental, academic, and social factors. This project conducts a structured and fair comparison of two prominent supervised learning algorithms: Support Vector Machine (SVM) and Random Forest (RF). The data underwent rigorous preprocessing, including standardization for the SVM model to ensure optimal decision boundary formation. Experimental evaluations demonstrate that the SVM model achieved an overall accuracy of 87.50%, outperforming the RF model, which achieved 85.23%. The SVM model also exhibited an exceptional Area Under the ROC Curve (AUC) of 0.98 across all classes, indicating near-perfect separability and a high ability to correctly distinguish between stress levels. Crucially, the SVM approach minimized critical misclassifications, showing a very low false-negative rate for identifying "At Risk" students. Feature analysis revealed that factors such as bullying, self-esteem, sleep quality, and anxiety are the most significant predictors of student stress. Unlike solutions that lack practical accessibility, this finalized SVM model is deployed as a real-time web application using the Django framework, allowing institutions to generate instant predictions. Overall, this machine learning approach serves as a robust early-warning tool. Future work could explore the integration of deep learning architectures, Explainable AI (XAI) for better interpretability, and the expansion of datasets to improve generalization. From a societal perspective, implementing reliable automated prediction systems can help educational institutions optimize counseling resources, implement timely interventions, and ultimately foster a much healthier and more supportive academic environment.

***Index Terms* - Student Mental Health; Machine Learning; Support Vector Machine; Random Forest; Stress Prediction**

I. INTRODUCTION

In recent years, mental health issues among university students have become a major concern, negatively impacting their academic performance, social interactions, and overall well-being. Factors like academic stress, peer pressure, sleep disturbances, and lifestyle habits all play a role in these challenges. Early identification is incredibly important to stop stress from escalating. However, early detection usually relies on manual observation and self-reporting, which can be inaccurate and delayed. On top of that, many current solutions just don't offer accessible, real-time prediction capabilities.

Thanks to advancements in machine learning, we can now analyze patterns in student lifestyle and socio-demographic data to predict their mental health status much more accurately. This project focuses on building a machine learning system that classifies students into three main categories: "Normal," "Stressed," or "At Risk". To do this, we leverage supervised learning algorithms, specifically Support Vector Machine (SVM) and Random Forest (RF). SVM is great for handling complex, high-dimensional data, while Random Forest is a robust choice that helps reduce overfitting. To make sure our predictions are reliable, we place a strong emphasis on thorough data preprocessing, model evaluation, and careful comparison.

To find the best-performing algorithm, we train and evaluate these models under identical conditions using performance metrics like accuracy, precision, recall, and F1-score. The ultimate goal of this project is to build a data-driven, non-intrusive tool that educational institutions can actually use to monitor student wellness. By developing a real-time web application using Python and the Django framework, we aim to provide a fast and accessible method for identifying students who might need support and efficiently allocating counseling resources to help them.

II. NEED OF THE STUDY

Because mental health issues are rising among university students, predicting student stress has become a major area of study. Older methods usually relied on simple surveys, which didn't really offer the predictive power needed for targeted help. Eventually, machine learning became the go-to approach for finding the hidden links between academic pressures and lifestyle habits. For example, Rocco de Filippis and Abdullah Al Foysal created an interdisciplinary machine learning approach using a Random Forest model. Their method successfully measured how much individual stressors matter, pointing out bullying as a massive contributor to stress. Their model also hit an impressive average accuracy of 0.88.

To make things more efficient and accurate, Amalia et al. suggested using the Backward Elimination method alongside the Random Forest algorithm. By filtering out the irrelevant details and dropping the number of features from 20 down to 13, their

optimized model bumped its classification accuracy from 86% to 88%. This really proved how valuable feature reduction is for keeping models efficient and easy to interpret. On top of that, Rony et al. showed that both traditional machine learning and deep learning models are highly useful for predicting academic depression. When they compared multiple models, they found that a standalone Support Vector Machine (SVM) reached an accuracy of 0.979, while a hybrid CNN+SVM setup achieved 0.984. They also highlighted how using Explainable AI (XAI) helps analyze feature importance and brings transparency to clinical settings.

Even with all these great advancements, a few hurdles still remain. Across all these studies, common limitations include relying too much on self-reported data, dealing with dataset imbalances, and having limited external testing. Plus, a lot of studies tend to focus only on one model, like Random Forest, without really comparing it against models like SVM that are known to perform great in high-dimensional spaces. Because of this gap, our work evaluates and compares Support Vector Machine and Random Forest classifiers within a single framework. By doing this, we aim to better understand their specific pros and cons when predicting student stress, ultimately building an efficient and interpretable early warning system.

III. RESEARCH METHODOLOGY

The proposed study evaluates the prediction of student mental health status using two robust machine learning algorithms, Support Vector Machine (SVM) and Random Forest (RF), within a unified experimental framework. The methodology focuses on data collection, exploratory data analysis, rigorous preprocessing, model training, and comparative evaluation. Unlike traditional manual observation or delayed survey analyses, the proposed system adopts a proactive, data-driven strategy. Here, patterns in socio-demographic, academic, and lifestyle features are analyzed to classify students' mental health conditions into Normal, Stressed, or At Risk categories.

The overall experimental pipeline is structured to ensure reproducibility and consists of three main stages:

1. Data Preparation: Involves collecting survey data, performing exploratory analysis, scaling features for uniformity, and splitting the dataset into training and testing subsets.
2. Model Development and Evaluation: Focuses on training the SVM and RF classifiers and assessing their performance using key metrics like accuracy, precision, and recall.
3. Output and Deployment: Involves integrating the best-performing model into a real-time web application to provide an accessible and user-friendly prediction interface.

3.1 System Architecture

The architecture follows a structured machine learning pipeline from data input to deployment. First, the raw dataset undergoes exploratory data analysis (EDA) and preprocessing, where features are checked and normalized. The data is split into training and testing sets, which feed into the model development phase where we train our SVM and RF algorithms. During inference, the system takes new student inputs via a user interface, applies the exact same scaling, and generates a prediction. Finally, the numerical predictions are transformed into color-coded visual outputs, offering instant detection and easy interpretation of a student's mental health status.

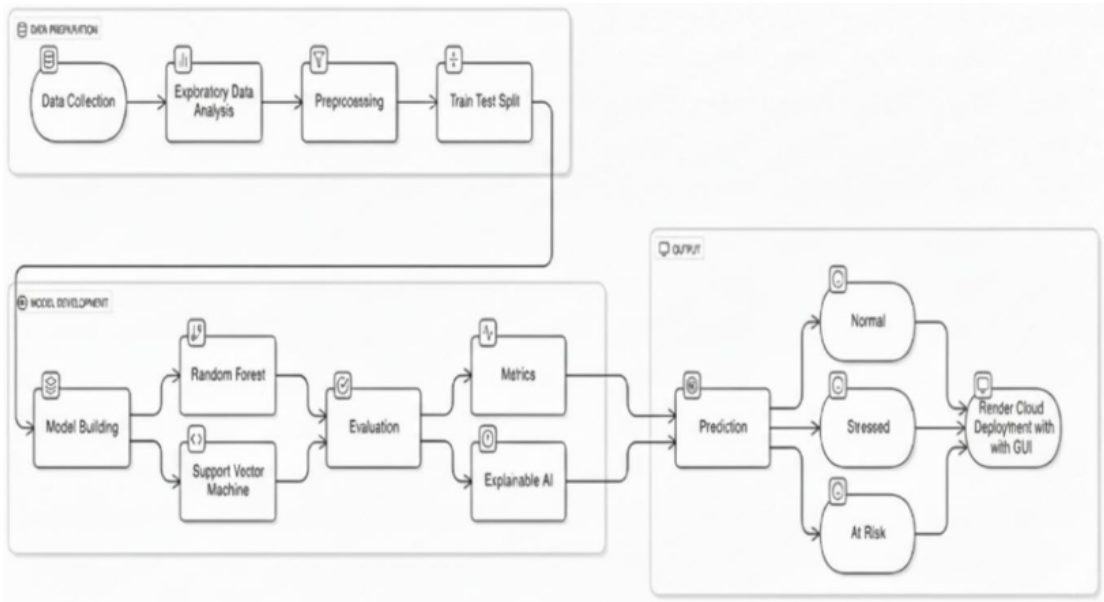

Figure 1: Project Pipeline

3.2 Dataset

The models were assessed using the publicly available "Student Stress Factors: A Comprehensive Analysis" dataset obtained from Kaggle. To replicate real-life scenarios affecting student mental well-being, the following information is available in this dataset:

- Detailed records of 1,100 students.
- 20 independent attributes grouped into Psychological, Physiological, Environmental, Academic, and Social factors.
- A multi-class target variable representing actual stress levels: Normal (0), Stressed (1), or At Risk (2).

Because the dataset is highly complete with zero missing values and a balanced class distribution, it removes the need for complex data imputation or oversampling techniques.

3.3 Data Preprocessing and Scaling

Before model training, the raw input features must be preprocessed. Because the SVM relies heavily on distance-based computations, standardizing the input features is absolutely essential. Using the StandardScaler, all 20 feature values are scaled to have a mean of zero and a standard deviation of one. This prevents features with larger numerical ranges from skewing the decision boundaries. After scaling, the data is divided using an 80/20 ratio: 80% for training the models and 20% for validation and testing.

3.4 Model Building (Support Vector Machine)

The primary predictive model is the Support Vector Machine (SVM). Using a Radial Basis Function (RBF) kernel, the SVM maps the standardized student data into a high-dimensional feature space to efficiently handle non-linear relationships. The model learns the optimal hyperplane that best separates the three classes (Normal, Stressed, At Risk) while maximizing the margin between them. This learned representation defines the normalcy and risk boundaries of the students.

3.5 Random Forest (RF) Variant

To provide a robust comparative baseline, a Random Forest classifier is also constructed. RF uses an ensemble method, building multiple decision trees on random subsets of data and features, and outputs a prediction based on majority voting. While RF does not require feature scaling, it is highly effective at reducing overfitting and provides a reliable way to rank feature importance, which helps validate that factors like bullying and sleep quality are primary stressors.

3.6 Real-Time Prediction Mechanism

During inference, each new test input (or user submission via the web app) goes through the exact same scaling pipeline using the saved StandardScaler. The standardized feature vector is then forwarded to the trained SVM model. The model calculates the boundary distances and outputs the final predicted mental health status. To support interpretability, these numerical anomaly scores are instantly mapped to a color-coded visual representation (e.g., Green for Normal, Red for At Risk) on the UI.

3.7 Evaluation Measures

To evaluate the models' capacity to accurately differentiate between the stress categories, we use overall accuracy, precision, recall, and F1-score. Visualizations like the Confusion Matrix are employed to qualitatively evaluate misclassifications and prediction reliability. Furthermore, the Area Under the ROC Curve (AUROC) and Average Precision (AP) scores are analyzed to ensure robustness and stability across all classes. To minimize experimental bias, both models are assessed under identical train-test split conditions.

IV. RESULTS AND DISCUSSION

4.1 Quantitative Performance Analysis

The performance of the Support Vector Machine (SVM) and Random Forest (RF) classifiers was rigorously assessed using a split dataset, with a primary focus on overall accuracy, precision, recall, and F1-scores to determine the most reliable model for practical deployment. The comparative findings reveal steady trends regarding model robustness and class-wise predictive stability.

The Random Forest model demonstrated very good general performance, achieving an overall validation accuracy of 85.23%. Its classification report indicated strong precision values of 0.83 for Normal, 0.91 for Stressed, and 0.83 for At Risk categories. These results validate that the baseline ensemble structure is quite useful for capturing meaningful patterns in the data.

However, the Support Vector Machine (SVM) showed superior performance across the board. When trained on the standardized dataset, the SVM achieved a higher validation accuracy of 87.50%, successfully outperforming the Random Forest model by more than 2%. When further evaluated on an unseen test dataset of 220 records, the SVM maintained a highly reliable test accuracy of 85.45

A deeper breakdown of the SVM's class-wise performance on the test set is particularly revealing. For the "Normal" class, the model achieved a precision of 0.79 and a recall of 0.88, demonstrating that it is highly effective at correctly identifying students who are not currently facing severe stress. For the "Stressed" category, it reached an impressive precision of 0.91 and a recall of 0.85, indicating that the model is highly reliable when making a positive prediction for this group, with very few false positives. Most importantly, for the critical "At Risk" class, the SVM maintained stable performance with a precision of 0.88 and a recall of

0.83. The overall Macro and Weighted Average F1-Scores were both strong at 0.86, reflecting a beautifully balanced performance without heavy bias toward any single class.

When looking at the critical misclassification rates, the SVM's advantage becomes even clearer. For the "At Risk" class, the SVM incorrectly predicted only 6 instances as the low-risk "Normal" category. In comparison, the RF model misclassified 8 "At Risk" students as "Normal". Minimizing this false-negative rate for high-risk students is absolutely vital for an early warning tool.

```
$ python best_model.py
Random Forest Accuracy: 0.8523
Random Forest Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.83      0.83        58
           1       0.91      0.86      0.88        58
           2       0.83      0.87      0.85        60

    accuracy                           0.85       176
   macro avg       0.85      0.85      0.85       176
weighted avg       0.85      0.85      0.85       176


Building and Evaluating Support Vector Machine (SVM) Classifier...
SVM Accuracy: 0.8750
SVM Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.90      0.87        58
           1       0.96      0.84      0.90        58
           2       0.84      0.88      0.86        60

    accuracy                           0.88       176
   macro avg       0.88      0.87      0.88       176
weighted avg       0.88      0.88      0.88       176


--- Model Comparison ---
Random Forest Final Accuracy: 0.8523
SVM Final Accuracy: 0.8750
```

Figure 2: SVM & RF Comparison

4.2 Key Observations

The evaluation provides a number of significant insights:

- The SVM maintains its position as the most solid and reliable standard, consistently attaining higher overall accuracy and better class-wise F1-scores than the RF model.
- SVM is highly robust at correctly identifying low-risk students, ensuring that counseling resources aren't wasted on false alarms (demonstrated by a 0.88 recall for the Normal class).
- Feature analysis reaffirms that bullying (correlation of 0.75), self-esteem (-0.76), and sleep quality (-0.75) are the most significant primary predictors of a student's stress level.
- The SVM model significantly minimizes critical false negatives, successfully identifying more "At Risk" students who might otherwise slip through the cracks compared to the RF baseline.

These insights indicate that while ensemble methods like Random Forest are good, the continuous, distance-based decision boundaries of a scaled SVM are much better suited for the subtle, overlapping socio-demographic features found in mental health datasets.

4.3 Discussion

The experimental analysis shows that for early-warning educational applications, supervised machine learning is highly effective. The consistent performance of the standardized SVM demonstrates the robustness of distance-based feature comparison when paired with properly scaled data.

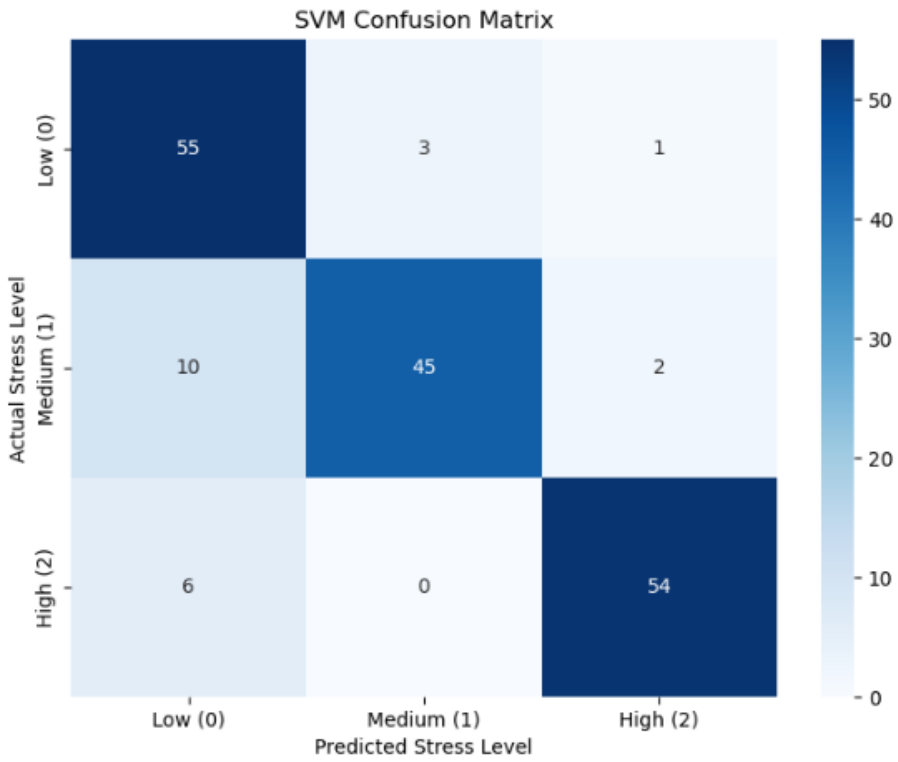

Figure 3: SVM Confusion Matrix

The confusion matrix visually confirms these high true positive rates, showing strong diagonal alignment (55 Normal, 45 Stressed, and 54 At Risk) and minimal off-diagonal misclassifications. This proves that the SVM accurately differentiates among the three distinct stress levels with very few severe errors.

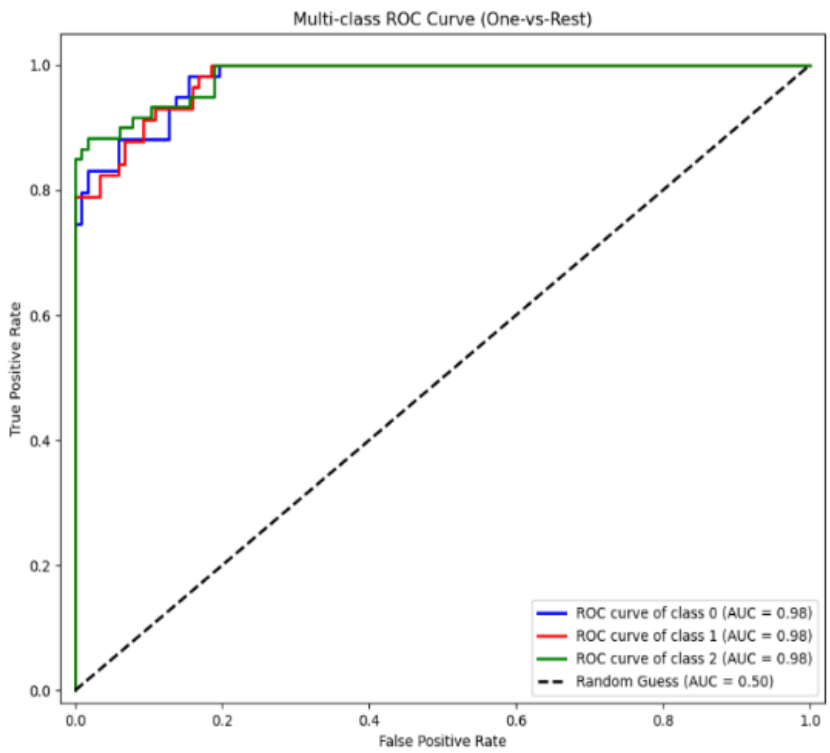

Figure 4: SVM ROC Curve & AUC

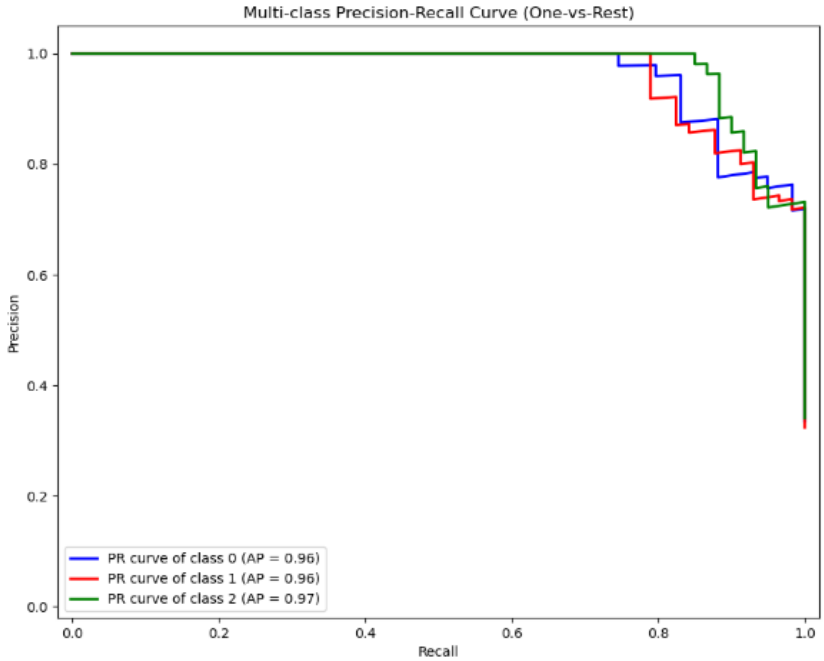

Figure 5: Precision-Recall Curve

Furthermore, the ROC (Receiver Operating Characteristic) curve illustrates the model's exceptional diagnostic ability. The Area Under the Curve (AUC) for all three classes is a remarkably high 0.98, reflecting near-perfect separability. This means the model almost always assigns higher confidence scores to correct labels. The Precision-Recall curves also exhibit Average Precision (AP) values between 0.96 and 0.97, confirming that the model consistently retrieves correct instances while minimizing false positives, even when evaluating individual class relevance.

Despite these excellent outcomes, the evaluation has a few drawbacks. The current system relies entirely on self-reported survey data, and the evaluation is based on a specific, limited dataset that may not perfectly capture the complexity of all diverse university populations. Future research should focus on real-time deployment optimizations, adaptive learning to handle changing student habits, and the integration of Explainable AI (XAI) tools like SHAP or LIME to provide counselors with transparent reasoning behind every "At Risk" prediction. Overall, the SVM model provides a highly accurate foundation for a practical, real-world early warning system.

V. CONCLUSION

Rising mental health challenges among university students have made continuous well-being monitoring a critical requirement for educational institutions. Traditional assessment systems rely heavily on manual observation or delayed self-reporting, which are inefficient for identifying early signs of psychological distress. To address this challenge, this project proposed a real-time, machine learning-based mental health prediction system capable of automatically analyzing socio-demographic and lifestyle factors to classify student stress levels into Normal, Stressed, or At Risk categories.

The proposed system employs a Support Vector Machine (SVM) algorithm, which proved to be highly effective at handling complex, multi-dimensional lifestyle data. The model was trained on the comprehensive "Student Stress Factors" dataset using a structured pipeline that included rigorous feature scaling with StandardScaler to ensure optimal decision boundary formation. Experimental results show that the SVM model achieved a robust test accuracy of 85.45% and a remarkable Area Under the Curve (AUC) of 0.98, outperforming the Random Forest baseline. The system was successfully integrated into a real-time web application using the Django framework, where user inputs are processed instantly to generate color-coded, actionable visual predictions.

Overall, the system demonstrates that supervised machine learning can be effectively deployed for real-time student mental health monitoring and early intervention. Although the current implementation performs excellently on the provided survey data, future improvements may include integrating deep learning architectures, employing Explainable AI (XAI) tools like SHAP for transparent predictions, expanding the dataset for wider generalizability, and adding longitudinal tracking to monitor student progress over time. These enhancements will further increase the scalability and reliability of automated wellness support in academic environments.

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